Constructions (Set B) — MCQ Worksheet

Mathematics • Chapter 11 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. To divide a line segment in ratio 5:7, the total number of points to be marked on the ray is:

(A) 5
(B) 7
(C) 12
(D) 35

Q2. To construct triangle similar to a given triangle with scale factor 2/3, the larger of m and n is:

(A) 2
(B) 3
(C) 5
(D) 6

Q3. To construct a tangent to a circle from external point P, an auxiliary circle is drawn with:

(A) OP as diameter
(B) Radius of given circle
(C) Any random radius
(D) OA as diameter

Q4. Number of tangents constructed from a point inside a circle is:

(A) 0
(B) 1
(C) 2
(D) Infinitely many

Q5. Scale factor for similar triangle construction with ratio 7/4 means new triangle is:

(A) Smaller
(B) Larger
(C) Same size
(D) Reversed

Q6. Justification of bisector of angle construction uses:

(A) SSS congruence
(B) SAS congruence
(C) AA similarity
(D) Pythagoras

Q7. To construct an angle of 60° we draw an arc and from there an arc of the same radius. The angle formed at the vertex is:

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Q8. Bisecting a 90° angle gives:

(A) 30°
(B) 45°
(C) 60°
(D) 75°

Q9. Perpendicular bisector of a segment is constructed using:

(A) Two arcs of equal radius from each endpoint
(B) Protractor
(C) Set square
(D) Straightedge only

Q10. Construction of triangle similar to given triangle with scale factor 1 produces:

(A) Larger
(B) Smaller
(C) Congruent triangle
(D) Mirror image

Q11. To construct tangent at a point on a circle, we draw a line perpendicular to:

(A) The chord
(B) The radius at that point
(C) The diameter only
(D) A tangent already drawn

Q12. Bisecting a 60° angle gives:

(A) 15°
(B) 30°
(C) 45°
(D) 20°

Q13. Equal-spacing while dividing a line uses the property:

(A) Basic Proportionality Theorem
(B) Pythagoras Theorem
(C) Mid-point theorem
(D) AA similarity

Q14. Construction of triangle similar with scale factor 3/5 produces a triangle of size:

(A) 3/5 of original
(B) 5/3 of original
(C) Equal to original
(D) Random

Q15. Why can we not construct a tangent from a point inside a circle?

(A) Tangents touch circle from outside
(B) Insufficient information
(C) Inside there is no point of contact
(D) Rule says so

Q16. To divide AB in ratio 3:5, we use a ray AX making acute angle with AB and mark 8 equal segments on AX. The dividing point on AB is joined to A_3 produced. Then in figure, point P on AB joining B and A_8 to A_3 is at:

(A) 3/8 of AB from A
(B) 5/8 of AB from A
(C) Midpoint
(D) 3/5 of AB from A

Q17. If scale factor for similar triangle is 4/5, sides of new triangle become:

(A) 4/5 of original sides
(B) 5/4 of original
(C) 1/2 of original
(D) Same

Q18. Constructing tangents from external point P to a circle of radius 3 cm where OP = 5 cm gives tangent length:

(A) 4 cm
(B) 5 cm
(C) 8 cm
(D) $\sqrt{34}$ cm

Q19. To construct a triangle similar to an equilateral triangle with each side 6 cm, scale factor 2/3, sides of new triangle are:

(A) 4 cm each
(B) 9 cm each
(C) 12 cm each
(D) 3 cm each

Q20. In construction of pair of tangents from external point, the midpoint M of OP is the centre of auxiliary circle whose radius is:

(A) OP/2
(B) OA
(C) Random
(D) OP

Q21. To construct a tangent from an external point P to a circle of radius r centred at O, the auxiliary circle passes through:

(A) Centre O only
(B) Point P only
(C) Both O and P
(D) Midpoint only

Q22. In constructing similar triangle to a given triangle ABC with scale factor 3/4, after marking 4 equal divisions on a ray from B, the third triangle is formed by joining:

(A) B_3 to A and C parallel correspondences
(B) B_4 to A
(C) Midpoints
(D) B_1 to A

Q23. Constructing an angle of 15° involves bisecting an angle of:

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Q24. Constructing an angle of 105° can be done by adding:

(A) 60° and 45°
(B) 60° and 60°
(C) 90° and 15°
(D) Both A and C

Q25. Tangent construction from external point P is justified because triangles OAP and OBP are:

(A) Congruent (RHS)
(B) Similar but not congruent
(C) Equilateral
(D) Not related

— End of Worksheet —